

Roll No. :

MID SEMESTER EXAMINATION MARCH 2025

Name of the Program: **B.Tech - CSE**

Semester: 8th

Name of the Course **Deep Neural Network based custom models and training**

Course Code: **TCS 836**

Time : 90 Minutes

Maximum Marks : 50

Note:

- i. Answer all the questions by choosing any one of the sub questions
- ii. Each questions carries 10 marks

Q1	(10 Marks)	CO2												
(a)	Discuss various types of production systems and their suitability to different AI problems.													
	OR													
(b)	State the bayes decision theory. How it works for 2-class category problem. Explain with suitable example.													
Q2	(10 Marks)	CO1												
(a)	Differentiate between keras and tensor flow library. Which one among both is more powerful to implement deep neural network. Justify your answer with example.													
	OR													
(b)	Elaborate the concept of feed forward and backward propagation in neural network. Write and discuss equation for optimize weight value for single perceptron.													
Q3	(10 Marks)	CO4												
(a)	How learning rate effects on performance of neural network. Discuss optimizer where learning rate is different for each neuron of each hidden layer in each epoch.													
	OR													
(b)	Discuss mini batch stochastic gradient optimizer. What is the disadvantage of this optimizer and how can we overcome it.													
Q4	(10 Marks)	CO3												
(a)	Differentiate between sequential and functional API.													
	OR													
(b)	In the table below, the x_i column shows scores on the aptitude test. Similarly, the y_i column shows statistics grades. Determine a) Regression Equation and b) Coefficient of determination: <table border="1" data-bbox="644 1537 1015 1771"><thead><tr><th>Student</th><th>x_i</th><th>y_i</th></tr></thead><tbody><tr><td>1</td><td>95</td><td>85</td></tr><tr><td>2</td><td>85</td><td>95</td></tr><tr><td>3</td><td>80</td><td>70</td></tr></tbody></table>	Student	x_i	y_i	1	95	85	2	85	95	3	80	70	
Student	x_i	y_i												
1	95	85												
2	85	95												
3	80	70												
Q5	(10 Marks)	CO3												
(a)	Differentiate between deep learning and machine learning.													
	OR													
(b)	What do you mean by hyperparameter tuning. How you can optimize neural network using hyperparameter tuning. Illustrate with suitable example.	CO2												

